

Ex. (i) In the number 16789352, the number formed by last 3 digits, namely 352 is divisible by 8.

$\therefore$ 16789352 is divisible by 8.

(ii) In the number 576484, the number formed by last 3 digits, namely 484 is not divisible by 8.

$\therefore$ 576484 is not divisible by 8.

6. Divisibility By 10:

A number is divisible by 10 only when its unit digit is 0.

Ex. (i) 7849320 is divisible by 10, since its unit digit is 0.

(ii) 678405 is not divisible by 10, since its unit digit is not 0.

7. Divisibility By 5:

A number is divisible by 5 only when its unit digit is 0 or 5.

Ex. (i) Each of the numbers 76895 and 68790 is divisible by 5.

8. Divisibility By 11:

A number is divisible by 11 if the difference between the sum of its digits at odd places and the sum of its digits at even places is either 0 or a number divisible by 11.

Ex. (i) Consider the number 29435417.

(Sum of its digits at odd places) – (Sum of its digits at even places)

$= (7 + 4 + 3 + 9) - (1 + 5 + 4 + 2) = (23 - 12) = 11$, which is divisible by 11.

$\therefore$ 29435417 is divisible by 11.

(ii) Consider the number 57463822.

(Sum of its digits at odd places) – (Sum of its digits at even places)

$= (2 + 8 + 6 + 7) - (2 + 3 + 4 + 5) = (23 - 14) = 9$, which is not divisible by 11.

$\therefore$ 57463822 is not divisible by 11.

9. Divisibility By 25:

A number is divisible by 25 if the number formed by its last two digits is either 00 or divisible by 25.

Ex. (i) In the number 63875, the number formed by last 2 digits, namely 75 is divisible by 25.

$\therefore$ 63875 is divisible by 25.

(ii) In the number 96445, the number formed by last 2 digits, namely 45 is not divisible by 25.

$\therefore$ 96445 is not divisible by 25.

10. Divisibility By 7 or 13:

Divide the number into groups of 3 digits (starting from right) and find the difference between the sum of the numbers in odd and even places. If the difference is 0 or divisible by 7 or 13 (as the case may be), it is divisible by 7 or 13.

Ex. (i) $4537792 \rightarrow 4 / 537 / 792$

$(792 + 4) - 537 = 259$, which is divisible by 7 but not by 13.

$\therefore$ 4537792 is divisible by 7 and not by 13.

(ii) $579488 \rightarrow 579 / 488$

$579 - 488 = 91$, which is divisible by both 7 and 13.

$\therefore$ 579488 is divisible by both 7 and 13.

11. Divisibility By 16:

A number is divisible by 16, if the number formed by its last 4 digits is divisible by 16.

Ex. (i) In the number 463776, the number formed by last 4 digits, namely 3776, is divisible by 16.

$\therefore$ 463776 is divisible by 16.

(ii) In the number 895684, the number formed by last 4 digits, namely 5684, is not divisible by 16.

$\therefore$ 895684 is not divisible by 16.

12. Divisibility By 6: A number is divisible by 6, if it is divisible by both 2 and 3.

13. Divisibility By 12: A number is divisible by 12, if it is divisible by both 3 and 4.

14. Divisibility By 15: A number is divisible by 15, if it is divisible by both 3 and 5.

15. Divisibility By 18: A number is divisible by 18, if it is divisible by both 2 and 9.

16. Divisibility By 14: A number is divisible by 14, if it is divisible by both 2 and 7.

17. Divisibility By 24: A given number is divisible by 24, if it is divisible by both 3 and 8.

18. Divisibility By 40: A given number is divisible by 40, if it is divisible by both 5 and 8.

19. Divisibility By 80: A given number is divisible by 80, if it is divisible by both 5 and 16.